MTH2003 Tutorial Sheet 2

Please attempt all of the questions before your week 5 tutorial, so that you can contribute to and benefit from the discussion.

1. Consider the differential equation:
$$y'' - 6y' + 9y = 10\,\mathrm{e}^{3t}.$$

   [i] Find the complementary solution.
   [ii] Find a particular solution.
   [iii] Write down the general solution.

2. Prove that for any p and q
$$\begin{aligned} \mathrm{e}^p + \mathrm{e}^q &= 2\mathrm{e}^{(p+q)/2}\cosh[(p-q)/2], \\ \mathrm{e}^p - \mathrm{e}^q &= 2\mathrm{e}^{(p+q)/2}\sinh[(p-q)/2]\,. \end{aligned}$$
   Hence show that if a, b and c are constants that satisfy $b^2 > 4ac$ then any solution of the ODE
$$ay'' + by' + cy = 0$$
   can be written in the form $y(t) = \mathrm{e}^{\alpha t}(A\cosh\beta t + B\sinh\beta t)$. Find α and β in terms of a, b and c.

3. Consider again the equation from the Examples Sheet:
$$2t^2y'' + 3ty' - y = 0\,, \quad t > 0\,.$$
   On the Examples Sheet you were given the solution $y_1 = 1/t$ and you found a second solution y_2.
   (***a***) Write down Abel's identity for this equation. Verify that y_1 and the y_2 that you found satisfy this identity.
   (***b***) Now imagine that you had not found y_2, and use the knowledge of y_1 and Abel's identity to find the general solution of this equation (the answer should be equivalent to that you found before, of course.)

4. On the Examples Sheet you found the general solution of the following nonhomogeneous, second order, linear ODEs:
$$y'' + 6y' + 8y = 16e^{-2t}$$
   and
$$y'' + 6y' + 8y = e^{3t}\sin(2t)\,.$$
   Using those results, write down the general solution of the ODE
$$y'' + 6y' + 8y = e^{-2t} + 1537e^{3t}\sin(2t)\,.$$